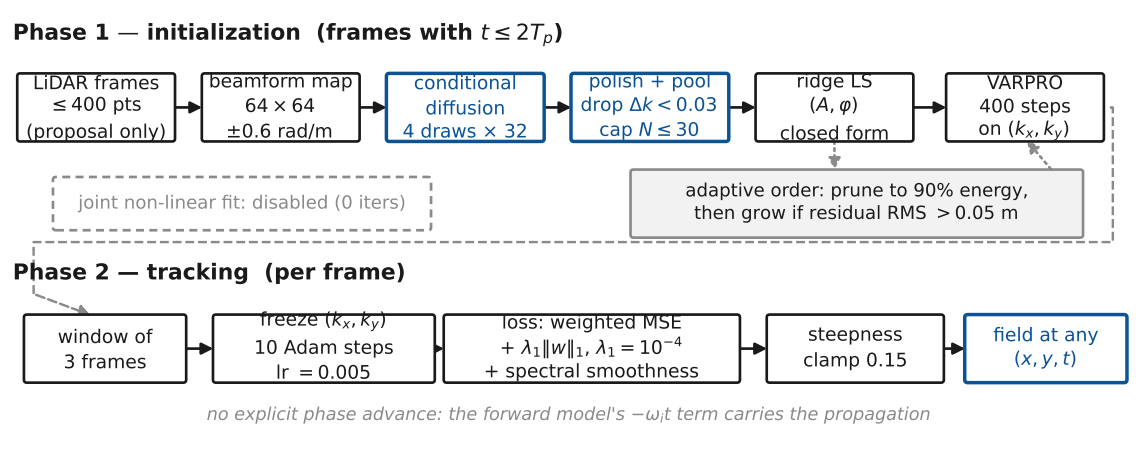
Phase 1 — initialization (frames with $t \le 2T_p$)
LiDAR frames ≤ 400 pts (proposal only)
beamform map 64 × 64 ±0.6 rad/m
conditional diffusion 4 draws × 32
polish + pool drop $\Delta k < 0.03$ cap $N \le 30$
ridge LS (A, φ) closed form
VARPRO 400 steps on (k_x, k_y)
joint non-linear fit: disabled (0 iters)
adaptive order: prune to 90% energy, then grow if residual RMS > 0.05 m
Phase 2 — tracking (per frame)
window of 3 frames
freeze (k_x, k_y) 10 Adam steps lr = 0.005
loss: weighted MSE + $\lambda_1 \|w\|_1$, $\lambda_1 = 10^{-4}$ + spectral smoothness
steepness clamp 0.15
field at any (x, y, t)
no explicit phase advance: the forward model's $-\omega_i t$ term carries the propagation